SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

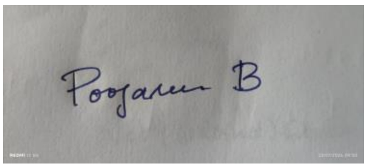
QUESTIONS: **Total Marks: 20**

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.

2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I POOJASREE B (192411091) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

INTERNET PROGRAMMING

COA-AT3

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1. Fundamental Concepts in Web Development

1. XML

XML is a markup language used to store, organise, and transport data in a structured format. It allows developers to define their own tags.

Purpose: XML is commonly used for data exchange between different applications and systems.

2. JSP

JSP is a server-side technology based on Java used to create dynamic web pages. It allows Java code and HTML to be combined to generate content dynamically.

Purpose: JSP is used to develop interactive web applications such as login pages, online shopping systems and DB driven websites.

3. MVC Architecture

MVC stands for Model-View-Controller. It divides a web application into three components.

Purpose: MVC makes applications easier to develop, maintain, test, and modify

4. XSLT

XSLT is a language used to transform XML documents into other formats, such as HTML, XML or text.

Purpose: It is mainly used to present XML data in a readable format on web pages.

5. XML Schema

XML Schema defines the structure, elements, attributes, and data types that are allowed in an XML document.

Purpose: XML Schema is used to validate XML documents and ensure that the data follows a predefined structure.

6. JSTL

JSTL is a collection of standard tags used in JSP pages to perform common tasks such as loops. conditions, formatting, and DB operations.

Purpose: JSTL reduces the need for Java code inside JSP pages and makes JSP applications easier to develop and maintain

7. PHP

PHP is a server side scripting language mainly used for developing dynamic web applications.

Purpose: PHP can process forms, manage sessions, interact with databases, and generate dynamic HTML pages.

8. DB Connectivity

DB connectivity refers to the process of connecting a web application to a db so that it can store. retrieve. update, and delete data.

Purpose: It allows web applications to work with user accounts, products, orders, student records, and other persistent data.

9. Parser

A parser is a software component that reads and analyzes structured data or Source code according to a defined syntax and converts it into a form that a program can process.

Purpose: Parsers are used to process XML, HTML, programming languages, and other structured documents.

2. Interconnection of Web Development Technologies

The given technologies work together to build a dynamic, structured, and data driven web application

1. XML

Stores and exchanges data in a structured format using user-defined tags.

2. XML Schema

Defines the structure and data types of XML and validates whether the XML data is correct.

3. Parser

Reads XML documents, checks their structure, and converts the data into a form the application can process.

4. JSP and PHP

Perform server-side processing. They receive user requests, process data, interact with databases, and generate dynamic web pages.

5. MVC Architecture

Organizes the application into:

Model - Handles data and db operations

View - Displays information to users.

Controller - Handles requests and connects Model and View.

6. JSTL

Provides predefined tags in JSP for loops, conditions, formatting, and other common tasks.

7. DB connectivity

Connects the web application with db's to perform CRUD operations such as creating, reading, updating and deleting data.

8. XSLT

Transforms XML data into formats such as HTML for presentation in a web browser.

Overall Flow

User

↓

JSP/PHP

↓

MVC

↓

Database/XML

↓

Processing

↓

JSP/XSLT

↓

Web page

Thus, these technologies work together to handle data representation, validation, processing, storage and presentation in a web application.